Statistics for AI

Lecture 3

Measures of Variability, Boxplot

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Measures of Variability

Measures of Variability

- A measure of location is only an incomplete description of the data. There is, of course, usually some variability in the data, so we need some summary statistics for that variability.

- Variability can be defined as the extent to which the values differ from each other. Other terms with very similar meanings are „diversity,“ „uncertainty“, „dispersion“, „spread“ or „variation“.

Did you hear about the statistician who put his head in the fridge and his feet in the oven?

Example

- We are looking at the long jump results of two people in meters:

A: 8.9   8.0   8.7   8.8   8.6   1.0

B: 5.0   4.3   4.2   4.7   4.1   4.0

- A first intuitive measure of variability is the range:

A: 7.9 m (8.9 − 1.0)

B: 1.0 m (5.0 − 4.0)

- A is obviously better and almost always has world record-breaking results. There is only one outlier at the last jump.

Simple Measures of Variability

- **Range = Maximum − Minimum**
  - sensitive for outliers
  - ignores all values between the smallest and the largest value
  - not an efficient measure for variability

➔ solution for the outlier-problem: exclude the smallest and largest 25% of the data

- **Interquartile-Range (IQR) = 3rd quartile − 1st quartile**
  - more robust measure of variability
  - still not an efficient method for measuring variability

Example: A (ordered): 1.0 8.0 8.6 8.7 8.8 8.9

➔ Range = 8.9 − 1.0 = 7.9

➔ IQR = 8.8 − 8.0 = 0.8

Variance

- Better idea: Measure the variability of the data as the averaged distances from each data point x_i to an appropriate location parameter
  - appropriate location parameter: arithmetic mean
  - measure for the distance: square of the distance

- **Variance for a population:**

$$Var(X) = \frac{1}{N} \sum_{i=1}^{N} \left(x_i - \bar{x}\right)^2$$

- The variance of a set of observations is the average of the squares of the deviations of those observations from their arithmetic mean.
- The variance will be large if the observations are widely spread around their mean and small if the observations are all close to their mean.

Example

- Back to A and B:

A	B
8.9	5.0
8.0	4.3
8.7	4.2
8.8	4.7
8.6	4.1
1.0	4.0

$$\bar{x}_A = \frac{1}{6}\left(8.9 + 8.0 + 8.7 + 8.8 + 8.6 + 1.0\right) = 7.333$$

$$\bar{x}_B = \frac{1}{6}\left(5.0 + 4.3 + 4.2 + 4.7 + 4.1 + 4.0\right) = 4.383$$

scale unit [meter]

$$Var(X_A) = \frac{1}{6}\left((8.9 - 7.333)^2 + (8.0 - 7.333)^2 + (8.7 - 7.333)^2 + (8.8 - 7.333)^2 + (8.6 - 7.333)^2 + (1.0 - 7.333)^2\right) = 8.106$$

$$Var(X_B) = \frac{1}{6}\left((5.0 - 4.383)^2 + (4.3 - 4.383)^2 + (4.2 - 4.383)^2 + (4.7 - 4.383)^2 + (4.1 - 4.383)^2 + (4.0 - 4.383)^2\right) = 0.125$$

scale unit [square-meter]

Example

- Due to a systematic measurement error (all distances were one meter too long) the mean and variance have to be calculated again.

A	B
7.9	4.0
7.0	3.3
7.7	3.2
7.8	3.7
7.6	3.1
0.0	3.0

$$\bar{x}_A = \frac{1}{6}\left(7.9 + 7.0 + 7.7 + 7.8 + 7.6 + 0.0\right) = 6.333$$

$$\bar{x}_B = \frac{1}{6}(4.0 + 3.3 + 3.2 + 3.7 + 3.1 + 3.0) = 3.383$$

The mean is reduced by one meter, just as expected.

$$Var(X_A) = \frac{1}{6}\left((7.9 - 6.333)^2 + (7.0 - 6.333)^2 + (7.7 - 6.333)^2 + (7.8 - 6.333)^2 + (7.6 - 6.333)^2 + (0.0 - 6.333)^2\right) = 8.106$$

$$Var(X_B) = \frac{1}{6}\left((3.0 - 3.383)^2 + (3.3 - 3.383)^2 + (3.2 - 3.383)^2 + (3.7 - 3.383)^2 + (3.1 - 3.383)^2 + (3.0 - 3.383)^2\right) = 0.125$$

The variance does not change (location-invariant).

➔ This is a good thing, because a measure of variability should not depend on the location of the data.

Example

- Out of curiosity we are multiplying the original values with 4. What happens to mean and variance?

A	B	4*A	4*B
8.9	5.0	35.6	20.0
8.0	4.3	32.0	17.2
8.7	4.2	34.8	16.8
8.8	4.7	35.2	18.8
8.6	4.1	34.4	16.4
1.0	4.0	4.0	16.0

$$\bar{x}_A = \frac{1}{6}\left(35.6 + 32.0 + 34.8 + 35.2 + 34.4 + 4.0\right) = 29.333$$

$$\bar{x}_B = \frac{1}{6}\left(20.0 + 17.2 + 16.8 + 18.8 + 16.4 + 16.0\right) = 17.533$$

The mean multiplies by the factor 4, just as expected.

$$Var(X_A) = \frac{1}{6}\left((35.6 - 29.333)^2 + (32.0 - 29.333)^2 + (34.8 - 29.333)^2 + (35.2 - 29.333)^2 + (34.4 - 29.333)^2 + (4.0 - 29.333)^2\right) = 129.689$$

$$Var(X_B) = \frac{1}{6}\left((20.0 - 17.533)^2 + (17.2 - 17.533)^2 + (16.8 - 17.533)^2 + (18.8 - 17.533)^2 + (16.4 - 17.533)^2 + (16.0 - 17.533)^2\right) = 1.996$$

The variance changes, but not by the factor 4.

Example

- Transformation properties of mean and variance:

	Mean A	Mean B	Conversion factor
starting value	7.333	4.383	
subtracting by 1	6.333	3.383	minus 1
multiplying by 4	29.333	17.533	times 4

➔ not location-invariant and not scale-invariant

	Variance A	Variance B	Conversion factor
starting value	8.106	0.125	
subtracting by 1	8.106	0.125	location-invariant
multiplying by 4	129.689	1.996	times $4^2 = 16$

➔ location-invariant but not scale-invariant

Standard Deviation

- Disadvantages of the variance are the „squared scale unit“ and the „squared scale transformation“.

- The solution to this problem is simple and leads to the so-called standard deviation as a measure of variability.

$$SD(x) = \sqrt{Var(x)} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(x_i - \bar{x}\right)^2}$$

- The standard deviation is the most common measure of variability and summarizes how far away from the arithmetic mean the observations are typically. It measures the degree of „randomness“ of objects around the arithmetic mean.

Calculating the Variance

- The formula for calculating the variance or the standard deviation is computationally intensive (first calculate each difference, then square these differences, then sum them up, and then finally divide this sum by N).
- An easier method of calculating the variance is based on Steiner’s theorem (named after Jakob Steiner, 1796-1863):

$$Var(X) = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - \bar{x}\right)^2 = \frac{1}{N}\sum_{i=1}^{N}x_i^2 - \bar{x}^2$$

$$Var(X) = \frac{1}{N}\left(\sum_{i=1}^{N}x_i^2 - \frac{\left(\sum_{i=1}^{N}x_i\right)^2}{N}\right)$$

Calculating the Variance

- Of course, we can use program packages to calculate the variance:

fx =VAR(B2:B6)

	A	B
1		**X**
2		3
3		4
4		7
5		12
6		9
7	**MEAN**	7
8	**VARIANCE**	13,5

Microsoft Excel

x

Mean	Variance	N
7,0000	13,500	5

SPSS

```r
x<-c(3,4,7,12,9)
mean(x)
## [1] 7
var(x)
## [1] 13.5
```

R

Using our previous formula, there is a huge difference in the results of the variance.

$$\bar{x} = \frac{1}{5}(3+4+7+12+9) = 7$$

$$Var(X) = \frac{1}{5}\left((3-7)^2 + (4-7)^2 + (12-7)^2 + (7-7)^2 + (9-7)^2\right) = 10.8$$

Sample Variance

- In our formula, we have assumed a population. This means that we are only interested in the data that we have collected or the data that is given. For this situation, the so-called „empirical variance of a population“ (= our formula) is appropriate.

- **Empirical Variance of a Population („Population Variance“):**

$$Var(X) = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - \bar{x}\right)^2$$

- If you want to estimate the „true variance“ of a population based on sample data, a different formula has to be used.

- **Sample Variance:**

$$Var(X) = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2$$

The sample variance is an unbiased estimator for the population variance.

Sample Variance vs. Population Variance

- **Why do we average by „n − 1“ in the case of a sample?**
  - Averaging by „n − 1“ results in an unbiased and consistent estimator for the variance of a population (details later).

- **Some remarks about the distances:**
  - The sum of the not squared distances $(x_i - \bar{x})$ is always zero.
  - That means the last distance can be found once we know the other „n − 1“ distances.
  - So we are not averaging n unrelated numbers. Only „n − 1“ of the squared distances can vary freely. Therefore, the number „n − 1“ is called the degrees of freedom of the variance and the standard deviation.

- The difference between the sample variance and the empirical variance becomes clearly very small with a large sample size.

- For now, we will use the formula for the empirical variance of a population in our examples. We will discuss this topic again in detail in the lecture covering „point estimators for population parameters“.

Notation

- To distinguish more clearly between population-setting and sample-setting the following notation is used.

- **Population of size N:**

  empirical mean: μ

  empirical variance: $Var(X) = \sigma^2$

  empirical standard deviation: $SD(X) = \sigma$

- **Sample of size n:**

  sample mean: $\bar{x}$

  sample variance: $Var(X) = s^2$

  sample standard deviation: $SD(X) = s$

The formula of the mean remains the same whether a population or a sample is given. Therefore often the notation $\bar{x}$ is used, regardless if it's a population or sample (details later).

Population Variance in Excel

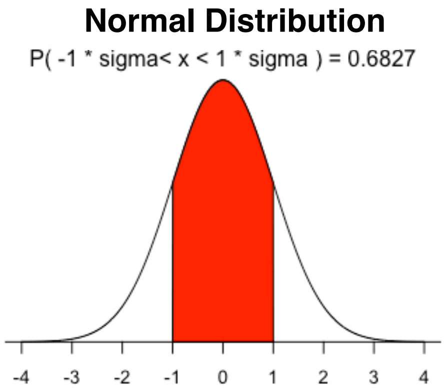
- In Excel, for example, the function „VAR. P“ can be used to calculate the empirical variance of a population.

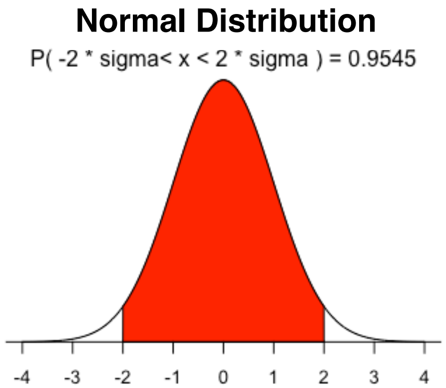
B8 =VAR.P(B2:B6)

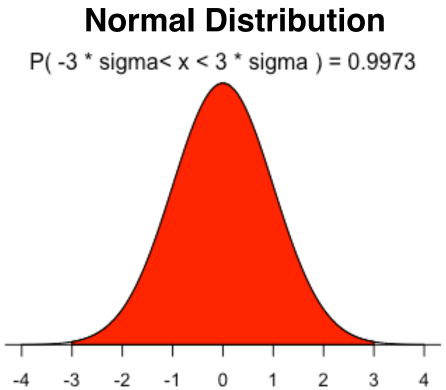
	A	B	C
1		**X**	
2		3	
3		4	
4		7	
5		12	
6		9	
7	**Function: MEAN**	**7**	mean
8	**Function: VAR.P**	**10,8**	population variance
9	**Function: VAR.S**	**13,5**	sample variance

Sigma Rule or „68–95–99.7 Rule“

- In the case of normally distributed data, a „sigma range“ can be defined based on the standard deviation.







The „1-sigma-range“ contains 68.269% of all data.

The „2-sigma-range" contains 95.450% of all data.

The „3-sigma-range" contains 99.730% of all data.

Variance in Case of Grouped Data

- In the case of grouped data, the squared distances are weighted based on the absolute frequencies f_i of the classes.

$$Var(X) = \frac{1}{N}\sum_{i=1}^{k} f_i\left(x_i - \bar{x}\right)^2 = \frac{1}{N}\sum_{i=1}^{k} f_i x_i^2 - \bar{x}^2$$

$$Var(X) = \frac{1}{N}\left(\sum_{i=1}^{k} f_i x_i^2 - \frac{\left(\sum_{i=1}^{k} f_i x_i\right)^2}{N}\right)$$

Example

- Based on the given frequency table of a continuous attribute, the variance and standard deviation should be calculated.

$[e_{i-1}, e_i)$	class mark x_i	class interval d_i	absolute frequency f_i	relative frequency p_i	f_i*x_i	$f_i*(x_i)^2$
4 up to 11	7.5	7	0	0.00	0	0
11 up to 18	14.5	7	12	0.12	174	2523
18 up to 25	21.5	7	24	0.24	516	11094
25 up to 32	28.5	7	28	0.28	798	22743
32 up to 39	35.5	7	24	0.24	852	30246
39 up to 46	42.5	7	12	0.12	510	21675
46 up to 53	49.5	7	0	0.00	0	0
		Total	**100**	**1.00**	**2850**	**88281**

$$\bar{x} = \frac{2850}{100} = 28.5$$

$$Var(X) = \frac{88281}{100} - 28.5^2 = 70.56$$

$$SD(X) = \sqrt{70.56} = 8.4$$

Coefficient of Variation

- If you want to compare the variation of different data sets with different scale units, you have to be careful, because variance and standard deviation depend on the scale unit of the data (e.g. data measured in [km] ➔ small values for variance; same data measured in [m] ➔ bigger values for variance, but variation is identical).

- A summary statistic which is independent of the scale unit and shows the variation in relation to the mean would be useful.

- **Coefficient of variation:**

$$C_V = \frac{SD(x)}{\bar{x}}$$

The mean body height of 10 people is 165 cm and standard deviation of the body height is 8 cm.

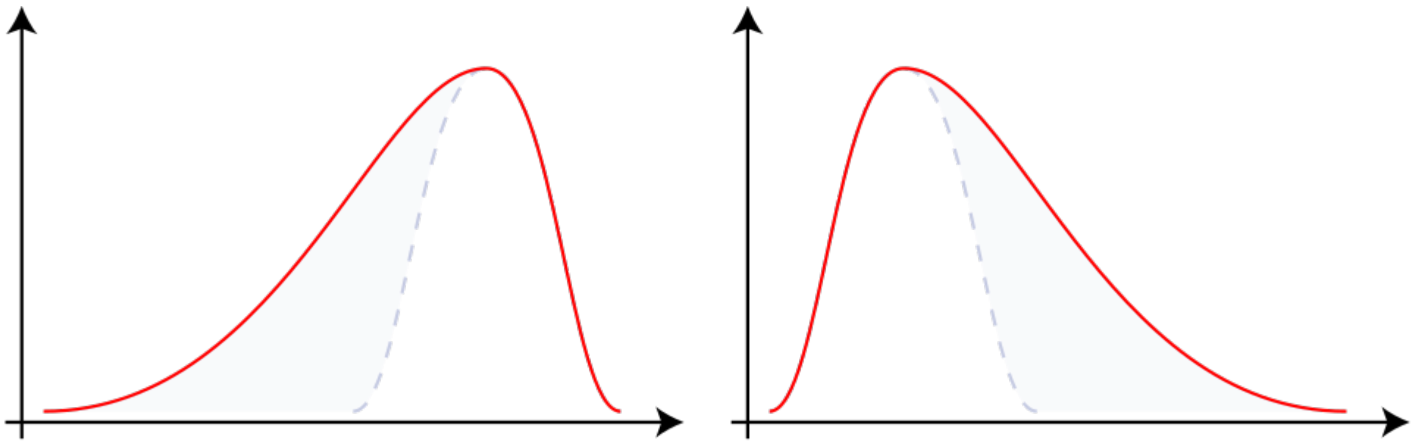
- Coefficient of variation: 8/165 = 0.048 = 4.8 %
- Variation of these persons is 4.8 % of the mean (independent of scale units)

Empirical Skewness

- The skewness is a measure of the asymmetry of a distribution:

$$\gamma = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \overline{x}}{SD(X)}\right)^3 \Rightarrow \begin{array}{l} < 0 \ \ldots \ \text{left skewed} \\ = 0 \ \ldots \ \text{symmetric} \\ > 0 \ \ldots \ \text{right skewed} \end{array}$$

(Empirical skewness for a population)



left skewed (negative skew) right skewed (positive skew)

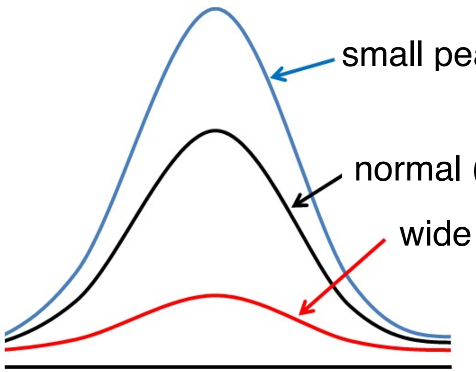
Empirical Kurtosis

- The kurtosis is a measure of the „peakedness“ of a distribution:

$$w = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \bar{x}}{SD(X)}\right)^4 - 3 \Rightarrow \begin{array}{l} < 0 \ldots \text{wide peaked} \\ = 0 \ldots \text{normal} \\ > 0 \ldots \text{small peaked} \end{array}$$

(Empirical kurtosis for a population respectively excess)

To be precise: The formula above represents an excess ➔ excess = kurtosis – „standardization parameter“
The „standardization parameter“ allows here an easy comparison with the normal distribution.

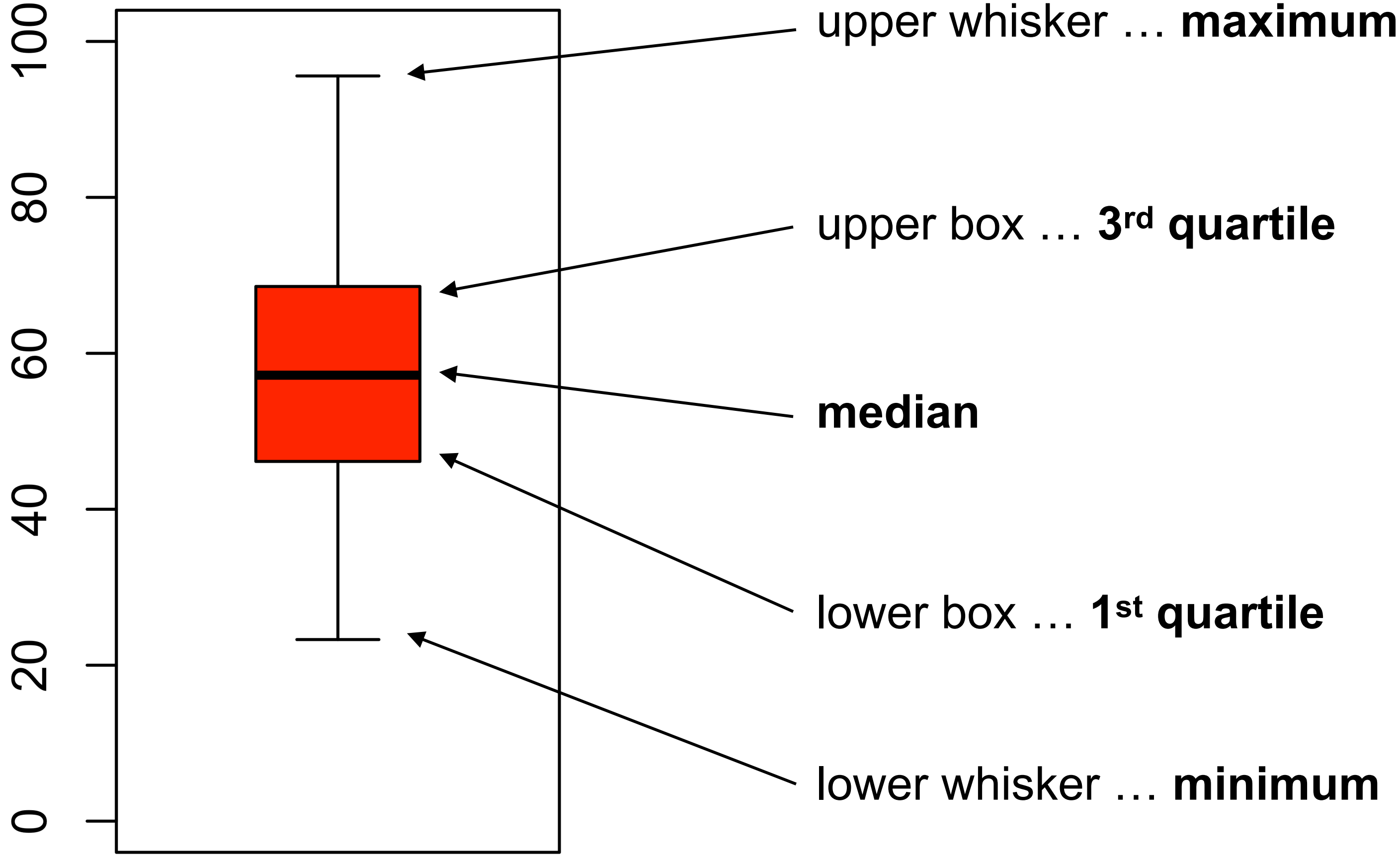


The idea of kurtosis as a measure of „peakedness“ goes back to Karl Pearson (1905). There is an ongoing discussion, if kurtosis really measures „peakedness“ („Kurtosis as Peakedness, 1905 – 2014. R.I.P“, Westfall P. Am. Stat., 2014) rather than „tailedness“ (extremity in terms of the tails of a distribution). My suggestion is, just use a histogram to get an idea of your data.

Boxplot

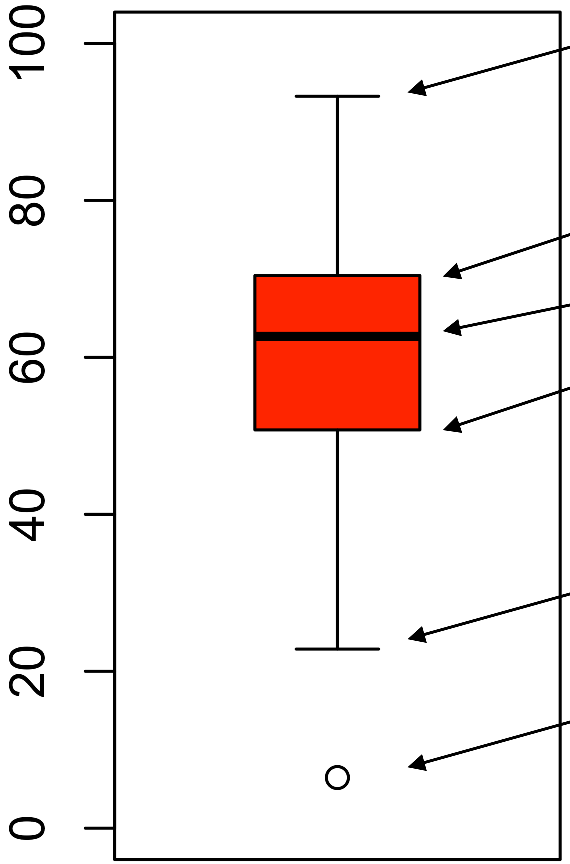
Boxplot (Box-Whisker-Plot)

- The boxplot is a graphical presentation of 5 summary statistics („five point summary“) and is based on the idea of John Wilder Tukey.



Boxplot (Box-Whisker-Plot)

- A modified boxplot allows easy detection of outliers.



upper whisker … highest value within „upper box + 1.5 · box-length“

upper box … 3rd quartile

median

lower box … 1st quartile

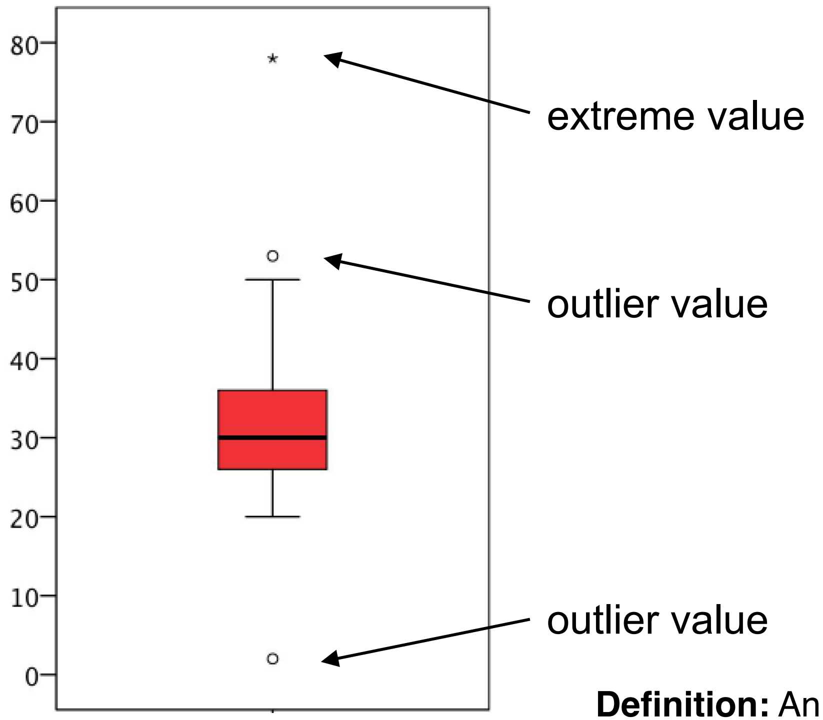
lower whisker … lowest value within „lower box – 1.5 · box-length“

outlier

Definition: An observation is called an outlier if it is more than 1.5 · IQR above the third quartile or below the first quartile.

Boxplot (Box-Whisker-Plot)

- Sometimes the outliers are distinguished between „outlier values“ and „extreme values“.
  - Outlier value (o) … under or over „box ± 1.5 · box-length (= 1.5 · IQR)“
  - Extreme value (*) … under or over „box ± 3 · box-length (= 3 · IQR)“
  -



Definition: An observation is called an outlier if it is more than 1.5 · IQR above the third quartile or below the first quartile.

An observation is called an extreme value if it is more than 3 · IQR above the third quartile or below the first quartile.

Example

- A boxplot of the age of 10 people should be created.
- Age in years: 45, 47, 50, 90, 51, 42, 49, 35, 31, 78

ordered age in years									
31	35	42	45	47	49	50	51	78	90
1	2	3	4	5	6	7	8	9	10

Quartiles according to Tukey:

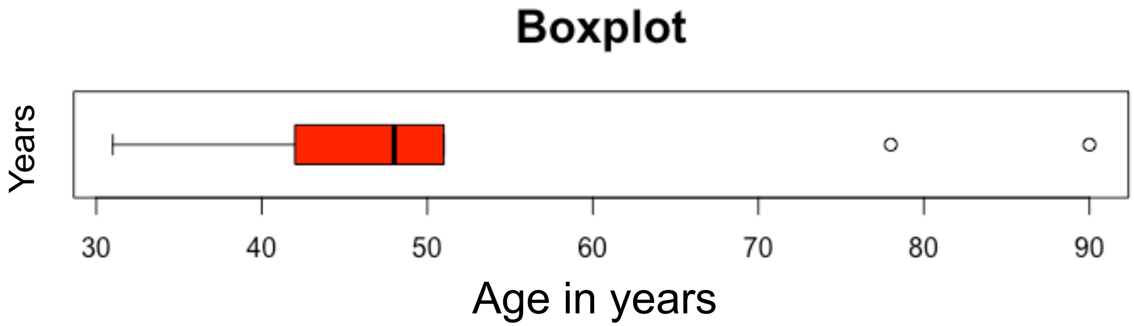
1^{st} quartile … 42 years

2^{nd} quartile … 48 years

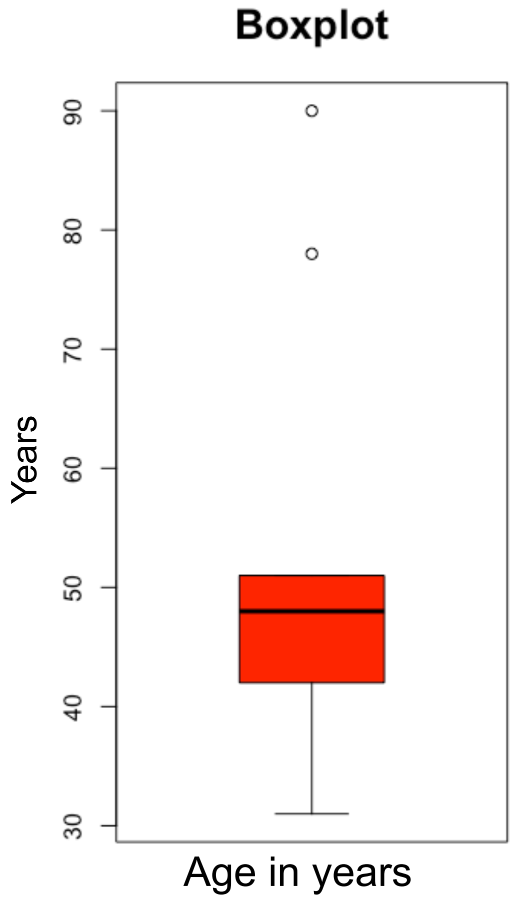
3^{rd} quartile … 51 years

Example

- Boxplot:



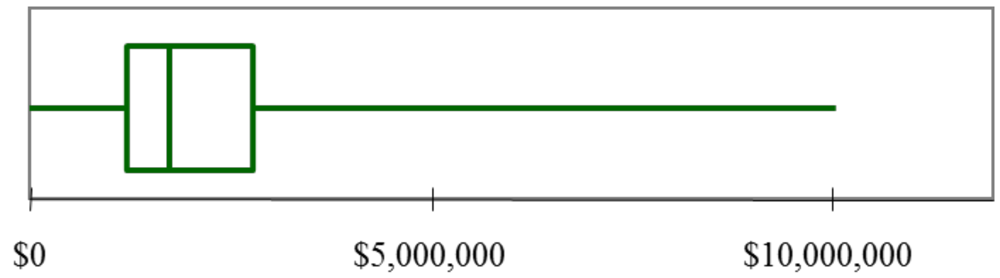
(horizontal boxplot)



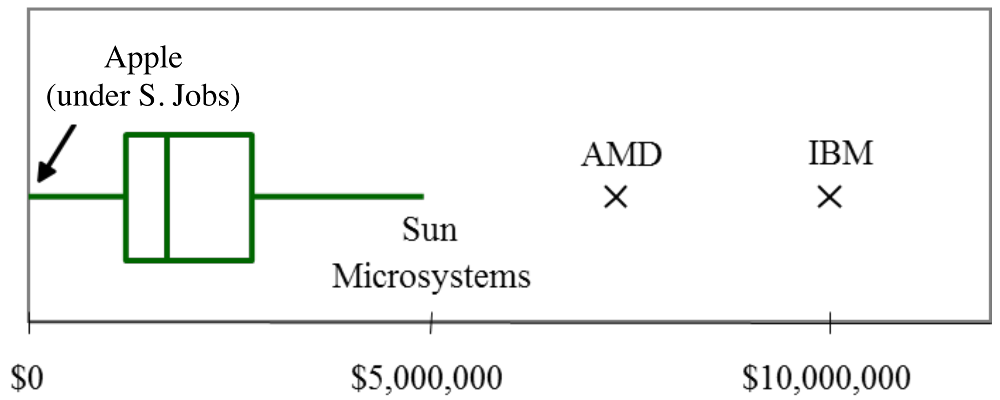
(vertical boxplot)

Example: CEO Income

- „standard“ boxplot:

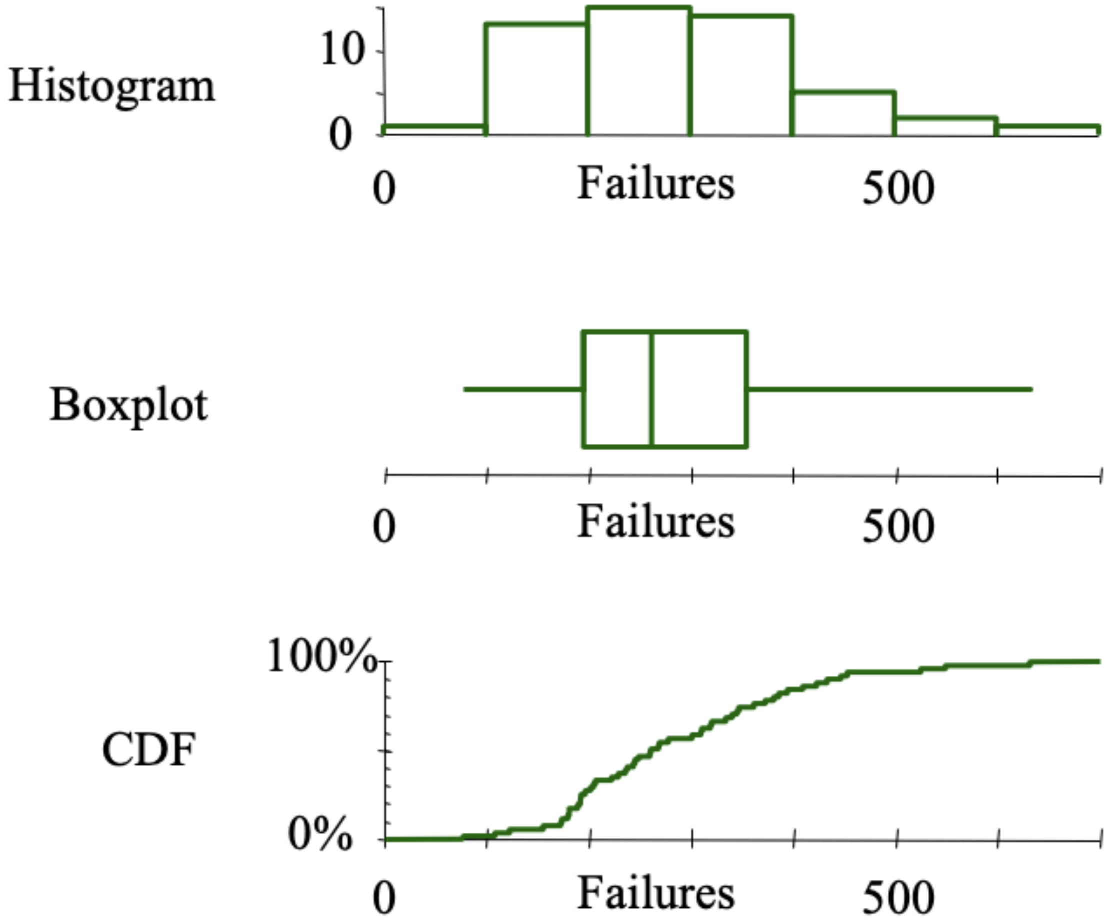


- „modified“ boxplot:

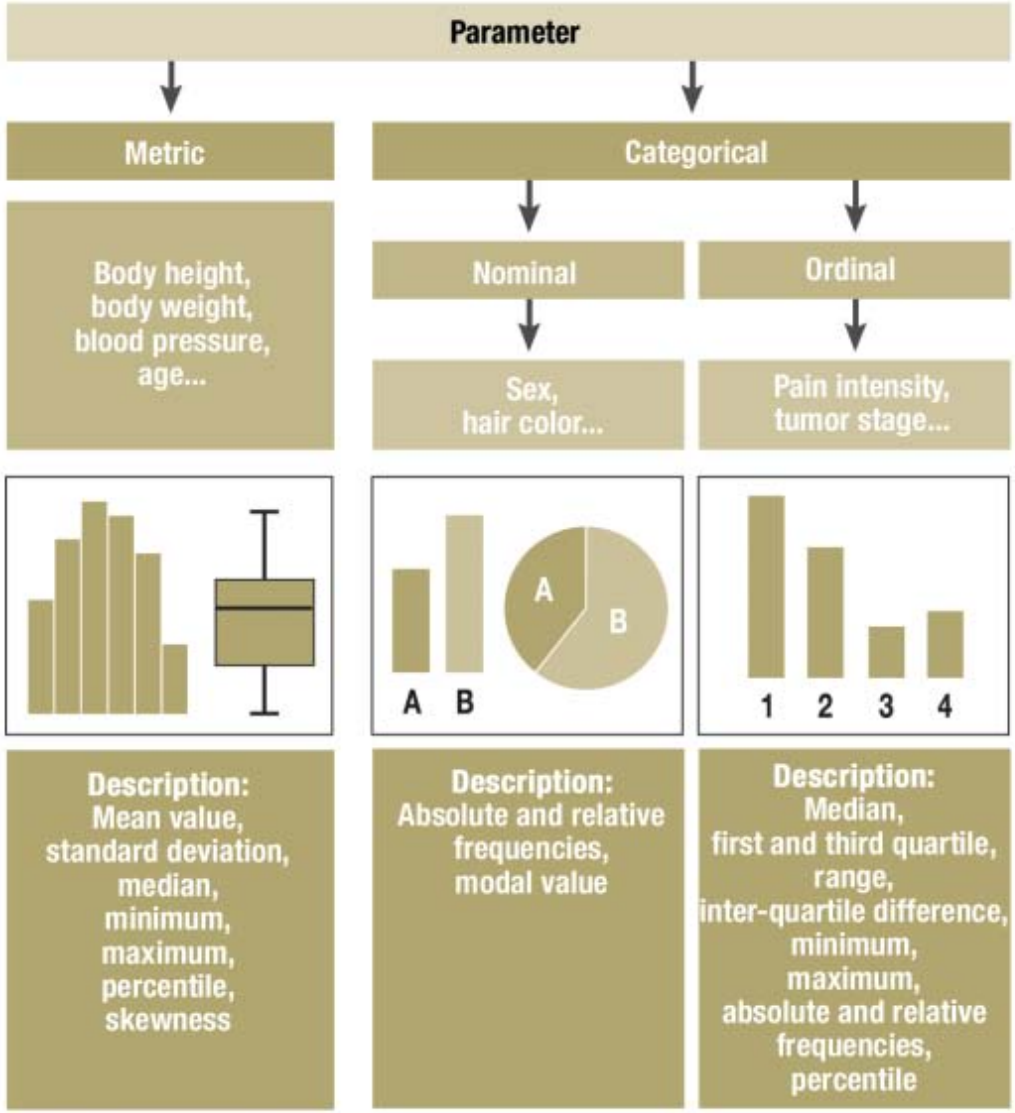


Example: Business Failure

- Graphical presentation of the number of business failures in various countries.



Summary



Dtsch Arztebl Int 2009; 106(36): 578–83